FOR COMPLETE BEGINNERS

Many values, one name.

Complete Modern Java, episode 14: arrays, in Java and TypeScript

> Read by index, change one value, and know the one number that makes Java stop.

Three prices are three variables. A hundred things would be a hundred variables.

An array keeps them under one name. Indexes start at 0, and the last is length - 1.

In this episode

- Declaring an array
- Indexes from 0
- length
- Changing one value
- Out of bounds
- Arrays.toString
- TypeScript arrays

Coming next

- Looping over an array (episode 15)
- Classes and objects (episode 16)

At the time of recording: Java 25, Node 24 LTS, IntelliJ IDEA 2026.1.3.

01 What each file printed

Prices.java: prices[0], prices[2], length	`150, 250, 3`
Prices.java: prices[1] = 120	`120`
Bounds.java: prices[3]	`ArrayIndexOutOfBoundsException: Index 3 out of bounds for length 3`
Show.java: IO.println(prices)	`a code such as [I@768b970c`
Show.java: Arrays.toString(prices)	`[150, 120, 250]`
prices.ts	`150, 3, undefined, [ 150, 120, 250 ]`

Every row was run, not predicted.

02 A row of numbered boxes

Several values of one type under one name. Index 0 is the first; the valid indexes run to length - 1.

A Java array has a fixed length: change what is in a box, never add a box.

Prices.java

```
void main() {
    int[] prices = {150, 99, 250};
    IO.println(prices[0]);        // 150
    IO.println(prices[2]);        // 250
    IO.println(prices.length);    // 3
    prices[1] = 120;
    IO.println(prices[1]);        // 120
}
```

> Printed 150, 250, 3, then 120.
>
> Run on a Mac, 2026-10-24, and again 2026-11-04: Java 25.0.4, TypeScript 7.0.2.

03 One past the end, and printing

prices[3] on three boxes stops the program. Read the message: index 3, length 3.

Printing an array directly shows a code; Arrays.toString lists the values.

Show.java

```
import java.util.Arrays;

void main() {
    int[] prices = {150, 120, 250};
    IO.println(prices);                    // [I@768b970c
    IO.println(Arrays.toString(prices));   // [150, 120, 250]
}
```

> Bounds.java: Index 3 out of bounds for length 3.
>
> Run on a Mac, 2026-10-24, and again 2026-11-04: Java 25.0.4, TypeScript 7.0.2.

04 TypeScript arrays

Past the end gives undefined, with no error. A JavaScript array can grow; const only stops replacing it.

prices.ts

```
const prices: number[] = [150, 99, 250];
console.log(prices[0]);        // 150
console.log(prices.length);    // 3
prices[1] = 120;
console.log(prices[3]);        // undefined
console.log(prices);           // [ 150, 120, 250 ]
```

> npx tsc --noEmit was quiet; node printed 150, 3, undefined, [150, 120, 250].
>
> Run on a Mac, 2026-10-24, and again 2026-11-04: Java 25.0.4, TypeScript 7.0.2.

Use an array

1. Many values of one type? One array, one name.
2. The first is index 0.
3. The last is length - 1.
4. To print it, use Arrays.toString.
5. Challenge: {100, 200, 300, 400}: prices[2] and the length.

Everything else

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The code and the challenge files: complete-modern-java
https://github.com/code-with-sam-dev/complete-modern-java

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